

Air University

Department of CyberSecurity NCSA



Business Intelligence and Management Integration

A Case Study of PIMS Hospital

Course: Introduction to Management
Final Term Project

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Executive Summary

This project analyzes the efficiency of operations and the digital transition of Pakistan Institute of Medical Sciences (PIMS) with the help of Business Intelligence (BI) and Management Integration. After critic review by Parliament, There has been a digital gap observed with only 6 out of 28 planned digital modules actually implemented and functional.

Our analyses uses Power BI dashboards to envision systemic bottlenecks, including a supply chain failure regarding Insulin and some vulnerabilities in the IT infrastructures. We implemented frameworks like SWOT Analysis, Porter's Five Forces and the BCG Matrix and PESTEL to encourage a transition towards a cloud-integrated, IoMT-secured future state to improve patient care and transparency in organization.

This case study demonstrates how BI tools can support decision-making processes related to management. It also helps in addressing key challenges in patient flow management. Inventory control and integration of digital systems by using Cybersecurity practices.

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1. Organizational Overview

1.1 Introduction to PIMS Hospital

PIMS Is one of the largest and recognized public sector tertiary care hospitals in Pakistan. It is located in G8 sector, Islamabad. It serves as national referral hospital and provides healthcare services to patients from twin cities to many remote regions across the country. It also plays an important role in administering specialized medical treatment, emergency services, and public healthcare support, particularly for low and middle income populations.

As it is federally administered healthcare institution, PIMS operates under the Ministry of National Health Services, Regulations and Coordination. Due to its high patient inflow and national importance, the hospital faces continuous pressure to maintain service quality, operational efficiency, and transparency in management practices.

1.2 Organizational Background

PIMS was established to provide affordable healthcare services. Being in the Capital also increased its accessibility. It also serves as a teaching and a research institution. Over time, the hospital has expanded greatly in terms of infrastructure, depts, and patient numbers. It currently consists of multiple clinical departments, diagnostic labs, many wards and some emergency units.

But despite the sheer scale and the importance, there were some parliamentary reports in recent past, specifically in December 2025. These reports have shown serious concerns regarding splitted digital systems, partial implementation of health information modules, and coordination challenges between IT teams and national health authorities encouraging digital transformation. Seeing this, PIMS has initiated projects to digitalize healthcare, only 6 out of 28 such modules are currently working. It shows limited digital implementation and low management visibility. Such challenges require PIMS to transition to Modern management practices and Business Intelligence (BI) solutions are required to transform operational data into good insights for decision makers.

1.3 Managerial Process and Administrative Structure

As this is federally administered healthcare institution that work under the Ministry of National Health Services, Regulations and Coordination. Basically, PIMS follows a formal bureaucratic and professional management hierarchy. The administration is designed to balance clinical excellence with public-sector accountability.

1.3.1 Administrative Hierarchy

The hospital is governed by a Board of Governors (BOG) and an the Management Committee, which provide strategic oversight and policy direction. The executive leadership consists of:

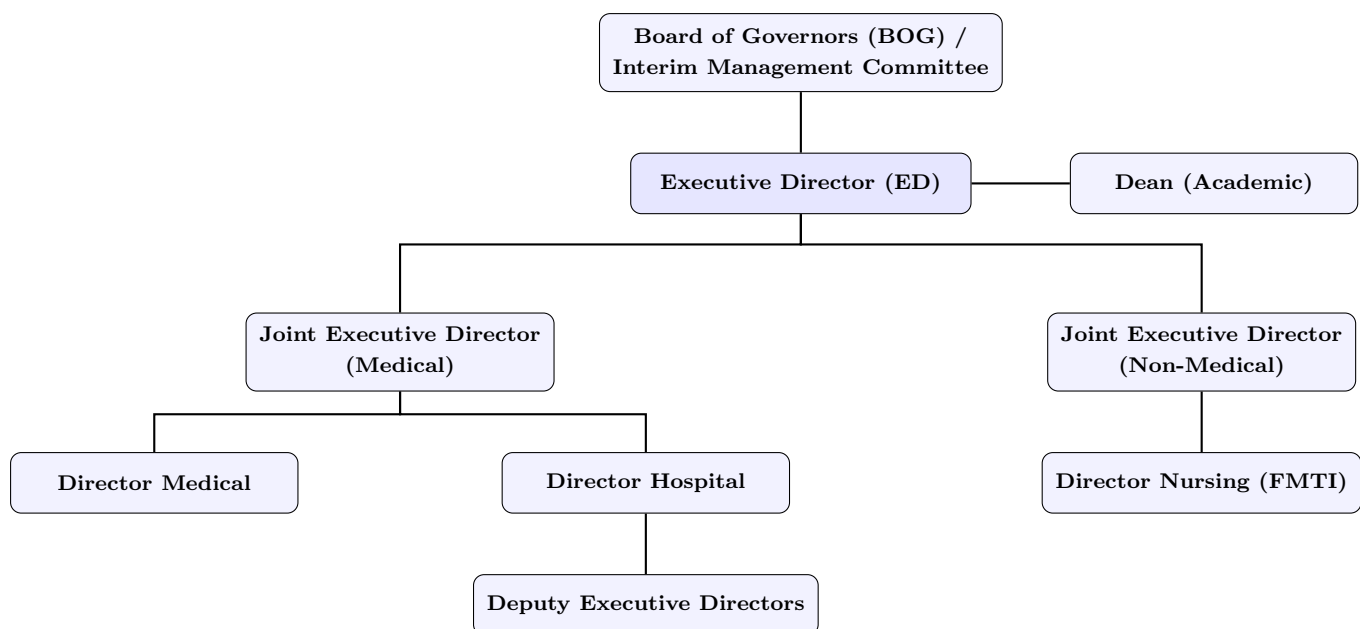


Figure 1.1: Administrative Hierarchy and Managerial Structure of PIMS

- **Executive Director (ED):** Basically, he is the overall administrative head, responsible for the entire institute's functioning and strategic alignment.
- **Dean:** Oversees academic and research standards, ensuring the hospital maintains its status as a teaching institution.
- **Joint Executive Directors (JED):** Basically, he will Divide Medical and Non-Medical roles to separate clinical oversight by using administrative logistics.
- **Directors of FMTI:** Specialized directors for Medical, Hospital, and Nursing wings ensure that each specialized workforce is managed by qualified professional staff.

1.3.2 Managerial Decision-Making and Control

Now for the Decision making process PIMS follow a centralized top-down approach for policy and budgeting, while clinical decentralization exists at the department level means the staff can recommend different decision at clinical department.

- **Strategic Level:** Now at this top level the BOG and ED set long-term goals, such as the digital transformation vision, aimed at streamlining regulatory processes.
- **Tactical Level:** Now this role is connected to the functional requirements where JEDs and Departmental Heads manage resource allocation, such as balancing staffing levels against bed occupancy rates.
- **Operational Level:** Medical and supporting staff execute daily patient care based on standardized clinical protocols.

Currently, PIMS utilizes *Reactive Control* (responding to issues like medicine shortages after they occur). Means when the problem happen like the insulin shortage or any other stockout then they react on it. But, Our proposed Business Intelligence integration introduces *Concurrent Control*, allowing real-time monitoring of hospital assets so they become pro-active to handle that stock out related problems.

1.4 Key Organizational Processes

Operations of PIMS revolve around many independent processes. For the purpose of this project, three core processes have been selected due to their direct impact on many important things including the safety of patients, efficiency of management and the delivery of service.

1.4.1 Patient Inflow and Emergency Management

Patient inflow management is one of the most critical operational processes at PIMS. Patients enter either through ER or OPD. At present, patient handling lies on manual registration modules and those also lack departmental integration. Due to the absence of an integrated digital patient ID system, tracking of patient movement and bed occupancy remains limited. This causes congestion during peak hours and delays during emergencies. If we look at this from a manager's view, no centralization of inflow data may make planning and resource allocation difficult.

1.4.2 Medicine Supply Chain and Inventory

The medicine supply chain at PIMS includes procurement, storage, and distribution through the hospital pharmacy. However, inventory tracking is not fully synchronized with consumption data. No tracking has led to stockouts of many essential drugs such as insulin, as already mentioned in the reports. The absence of predictive inventory mechanisms prevents proactive replenishment, exposing the hospital to patient safety risks. Inventory management should always be essential to make sure that there is uninterrupted patient care. Makes it an important area to focus for BI driven optimization.

1.4.3 IoMT Infrastructure and Cybersecurity

This process involves the integration of medical devices (ventilators, monitors) and security assets (CCTV, access control) into a secure digital network.

- **IoMT Integration:** Our analysis maps the transition where 66.67% of critical medical machinery is already IoMT-enabled.
- **Cybersecurity Focus:** The managerial challenge is to secure these nodes against data silos. By integrating Cybersecurity with Management, we ensure that the activation of the remaining 22 digital modules occurs within a secure, encrypted framework to protect patient privacy and system integrity and by applying cybersecurity in IoMT we can save patients lives from different cyber attack on the hospital machinery.

2. Process Analysis and Mapping

2.1 Patient Flow Process Mapping

The primary objective of mapping a process at PIMS is to evaluate how the institution utilizes its current infrastructure to manage high patient volumes. By mapping them, we can shortlist specific stages where BI and new technologies can be used to improve operational efficiency and informed decision making.

2.2 AS-IS Process Analysis (Current State)

The "AS-IS" state represents the existing operational reality at PIMS. While the initial administrative registration has been digitized, the subsequent clinical and ancillary workflows remain fragmented. This state relies heavily on the physical movement of documents and manual verification between departments, creating significant bottlenecks in patient care.

2.2.1 Professional Patient Flow Diagram (AS-IS)

The following flowchart illustrates the current fragmented patient journey. In this model, the patient essentially acts as the primary "data carrier" because clinical modules are not still implemented throughout hospital network.

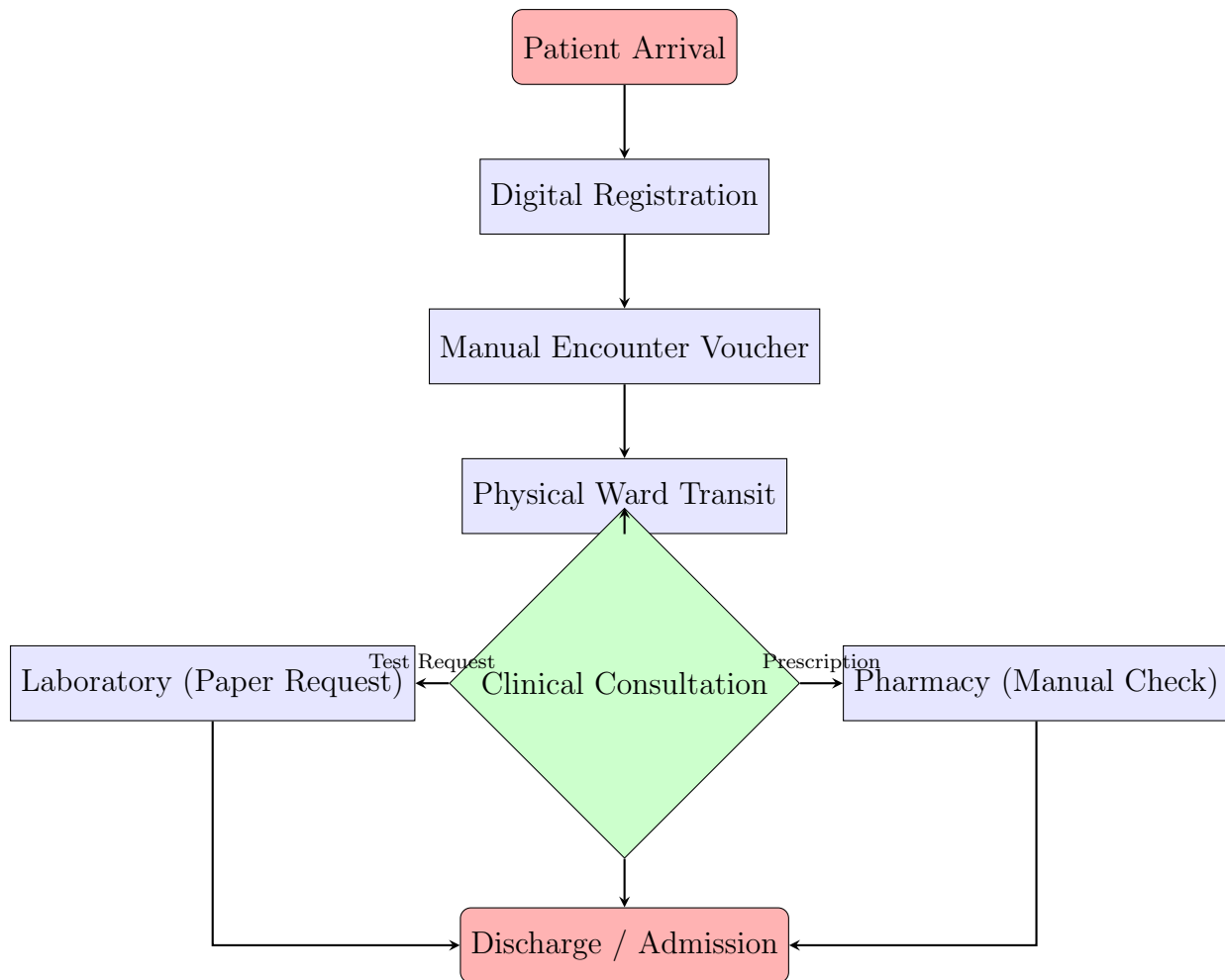


Figure 2.1: AS-IS Patient Flow: Fragmented Digital Integration at PIMS

2.2.2 Operational Breakdown of Challenges

- Initial Encounter Gaps:** Some gaps have existed in PIMS Hospital's operations from the beginning. For example, a patient's demographics are recorded when they arrive at the central desk, but this data silo is instantly created when the Manual Encounter Voucher is issued. It appears to be a major issue since it prevents real-time visibility for clinical information.
- Clinical Information Inconsistency:** Most of the time, doctors do not have complete access to digital patient histories. This results in the increase of workload and the need to rely only on verbal reports either from patients or just files. It seems ineffective and tiring.
- Supply Chain Disconnect:** Additionally, the pharmacy and consultation units are not connected in the supply chain. Stockouts result from the lack of a digital connection, particularly when it comes to frequently mentioned issues like insulin shortages.

2.3 TO-BE Process Design (Optimized State)

In order to accomplish the "One Patient, One ID" goal, the proposed "TO-BE" state envisions an integrated digital ecosystem using emerging technologies. across all 28 planned modules, this optimized model ensures that data automatically follows the patient.

- **Unified Electronic Health Records (EHR):** A centralized record that flows digitally between departments, eliminating the need for manual vouchers.
- **Automated Supply Chain Integration::** In order to prevent stock outs and initiate automated reorder points, pharmacy inventory and physician prescriptions are synchronized in real-time.

2.4 Challenges and Bottlenecks

Applying strategic management concepts to the PIMS workflow reveals several areas of inefficiency:

- **System Fragmentation:** There is a lot of disparity between 22 inactive and 6 active modules. So, End-to-end operational transparency is hindered by it.
- **Departmental Silos:** : Decision-making is less effective because pharmaceutical and diagnostic data are kept separate from the main record which is clinical.
- **Governance and Conflict:** Organizational friction between internal IT departments and external authorities stalls the necessary technical harmonization for project completion.

3. Business Intelligence and Emerging Technologies

3.1 Business Intelligence Tool Selection

For this study, Microsoft Power BI was chosen as the main Business Intelligence (BI) tool. From a management's view, it works as a Decision Support System that can combine chunks of hospital ward data into a single "Source of Truth." With this implementation, PIMS leadership can transform from reactive management to data based decision making, which is crucial for improving clinical results and organizational effectiveness.

3.2 Dataset Description

The analysis is based on authentic operational datasets modeled to reflect the hospital's current status and official scrutiny findings. The data includes:

- **Operational Metadata:** : Measures of waiting times, department referrals, and patient arrival times.
- **Supply Chain Logs:** Daily stock levels of critical, life-saving medicines such as Insulin.
- **Digital Infrastructure Status:** Real-time implementation tracking of the 28 digital health modules.
- **IoMT Connectivity Metrics:** Diagnostic data from medical machinery (ventilators, monitors) to evaluate equipment utilization.

3.3 BI Dashboard Analysis

To address the inefficiencies found in the "AS-IS" process mapping and offer complete management visibility, the dashboard is divided into four specialized views.

3.3.1 Hospital Operations Dashboard

This view provides **Operational Visibility** into the patient inflow process. By monitoring bed occupancy rates and ER congestion in real-time, management can identify systemic bottlenecks and reorganize staff allocation to handle peak loads effectively.

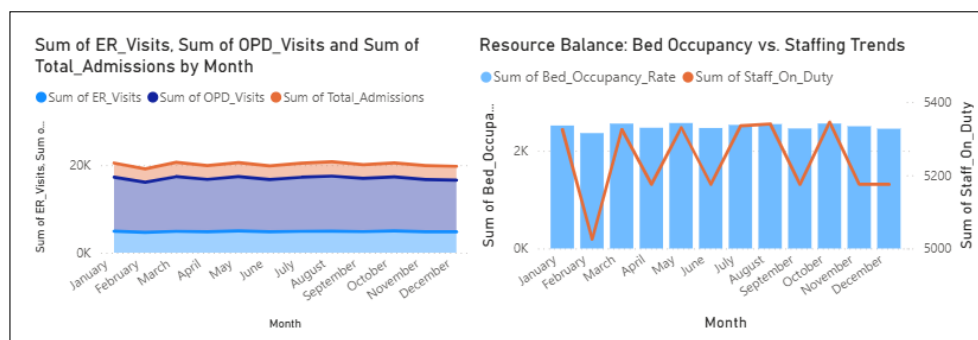


Figure 3.1: Operational View: Monitoring Patient Inflow and Resource Balancing

3.3.2 Inventory Management Dashboard

This dashboard serves as a **Control Mechanism** for the supply chain. It utilizes automated "Stock-out Alerts" to highlight critical shortages, specifically tracking Insulin inventory to mitigate the patient safety risks identified in recent parliamentary audits.

Medicine_Name	Sum of Stock_Quantity	Sum of Reorder_Level
Azithromycin 2200mg	892	78
Azithromycin 2300mg	447	73
Azithromycin 2400mg	1002	68
Azithromycin 2500mg	557	63
Azithromycin 2600mg	112	58
Azithromycin 2700mg	667	53
Azithromycin 2800mg	222	48
Azithromycin 2900mg	777	43
Azithromycin 3000mg	332	38
Azithromycin 3100mg	887	33
Azithromycin 3200mg	447	28
Total	275750	59350

Figure 3.2: Supply Chain View: Real-time Inventory Tracking and Predictive Alert System

3.3.3 IoMT Device Utilization Dashboard

By integrating data from the **Internet of Medical Things (IoMT)**, This dashboard tracks ventilator and lab equipment connectivity by using data from the Internet of Medical Things (IoMT). It helps with resources Optimization, making sure pricey medical equipment works, and accurately entering data into the patient's digital record

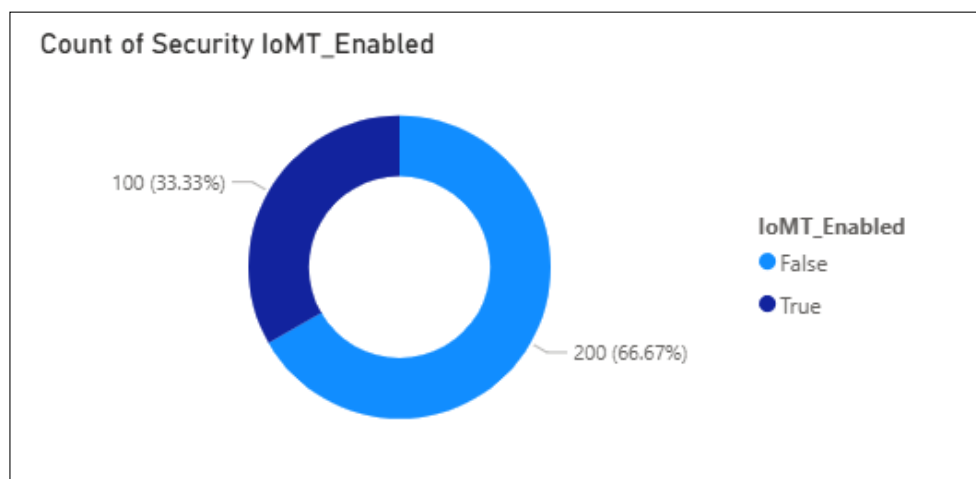


Figure 3.3: Technical View: Monitoring Medical Device Connectivity and Health

3.3.4 Risk and Vulnerability Dashboard

This section is a **Strategic Management Tool** that visualizes the "Digital Gap." By highlighting that only 6 of 28 modules are active, it identifies the high-level strategic risk caused by internal IT coordination challenges.

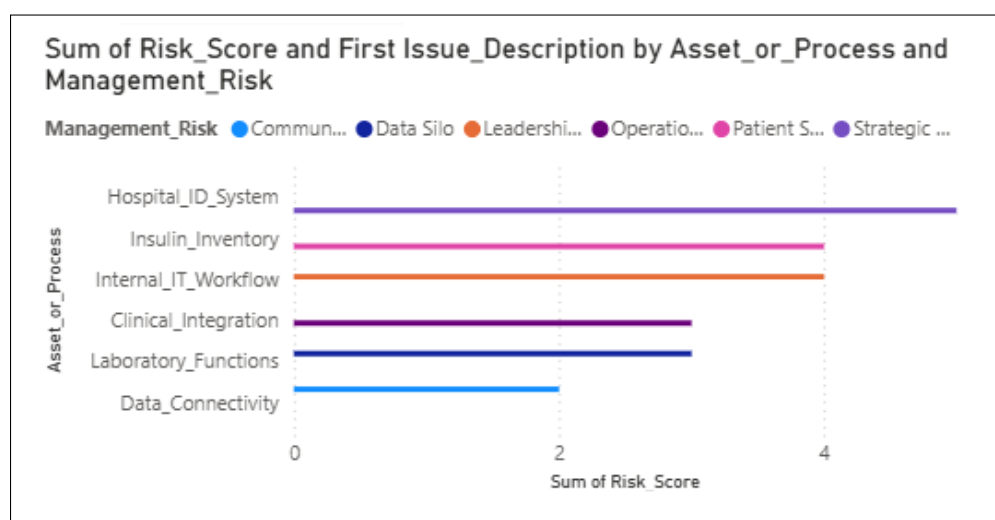


Figure 3.4: Strategic View: Digital Implementation Gaps and Operational Risks

3.4 Emerging Technology Recommendations

To facilitate the transition to the "TO-BE" state, the following technologies are recommended:

- **Artificial Intelligence (AI):** Implementation of AI for **Predictive Modeling** to forecast seasonal patient surges, allowing for proactive staff and resource planning and also for the vulnerabilities in the medical devices because these issue will lead to different health issue.
- **Cloud Computing:** Transitioning to a **Hybrid Cloud** environment to eliminate data silos and connect PIMS to national databases for universal patient identification.
- **Internet of Medical Things (IoMT):** Deploying smart sensors in storage areas to automate inventory tracking, thereby removing manual entry errors and ensuring the availability of critical drugs.

4. Application of Management Concepts

4.1 Strategic Analysis

Strategic planning at PIMS involves evaluating the organization's long-term objectives through structured management frameworks to enhance decision-making and operational effectiveness.

4.1.1 SWOT Analysis

The SWOT analysis provides a diagnostic view of PIMS's internal strengths and external pressures so the organization will check there different strengths and weaknesses:

- **Strengths:** Dominant market share in public healthcare and a large clinical data repository.
- **Weaknesses:** Serious implementation gaps (6/28 modules) and fragmented departmental communication and poor management of their inventory stocks.
- **Opportunities:** Federal support for "Universal Medical Records" and provincial system integration.
- **Threats:** Intensive parliamentary scrutiny and patient safety risks arising from supply chain failures and cyber related threat to data and IoMT (Medical devices).

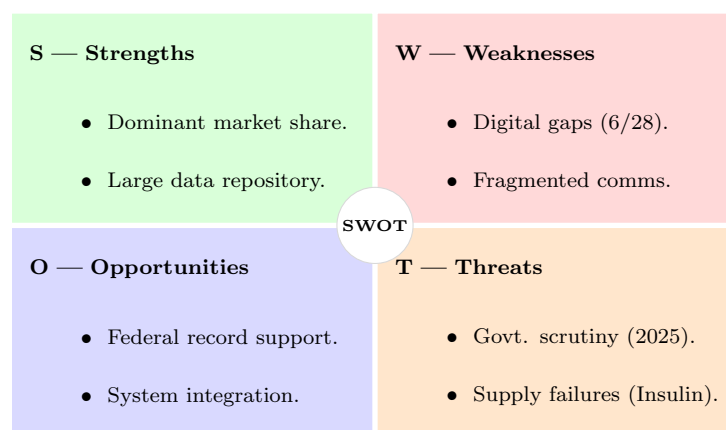


Figure 4.1: Condensed SWOT Analysis of PIMS Hospital

4.1.2 Porter's Five Forces

This framework analyzes the competitive forces within the healthcare sector:

- **Threat of New Entrants:** Low, due to massive capital requirements and regulatory barriers for tertiary institutions.
- **Bargaining Power of Suppliers:** High, specifically regarding IT vendors and pharmaceutical suppliers, as evidenced by the Insulin stockouts.
- **Bargaining Power of Buyers:** Moderate; while patients have limited public alternatives, their social and political influence triggers management audits.
- **Threat of Substitutes:** Moderate, coming from specialized private clinics for specific surgical procedures.
- **Intensity of Rivalry:** Moderate, centered on the competition for federal budget and specialized medical talent.

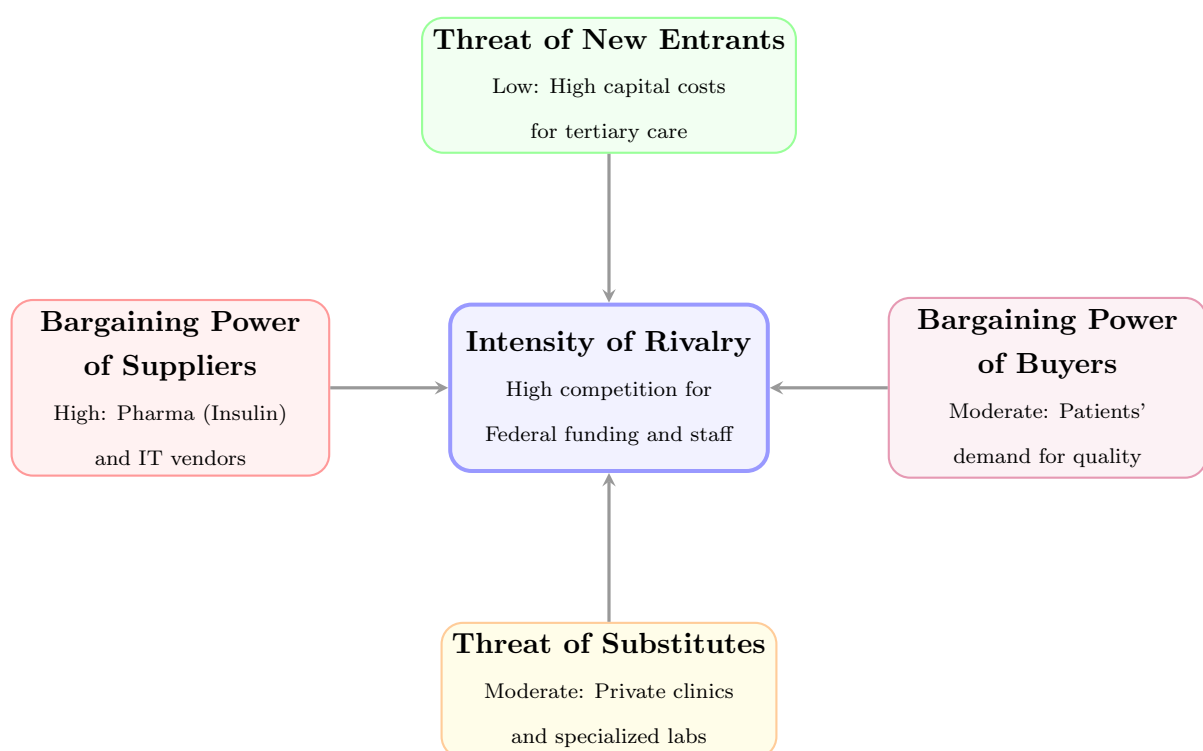


Figure 4.2: Porter's Five Forces Analysis for PIMS Hospital

4.1.3 BCG Matrix

Services are evaluated to determine where PIMS should prioritize its limited digital budget:

- **Stars:** Emergency and ICU services (High growth, High demand).
- **Cash Cows:** Outpatient Department (OPD) and Diagnostic Labs (Stable volume).
- **Question Marks:** The Digital Transformation Project (High potential, currently stalled at 6 modules).
- **Dogs:** Manual, paper-intensive administrative workflows (Low efficiency).

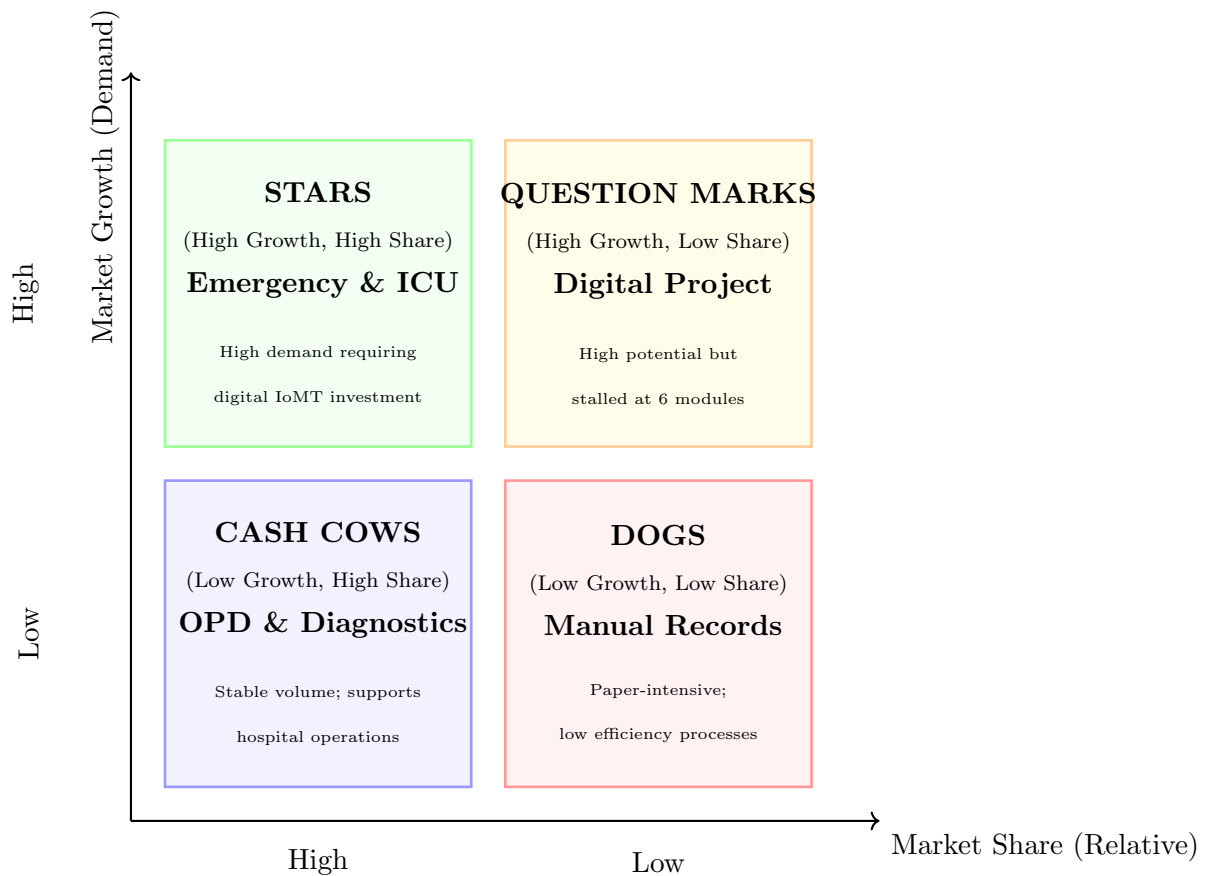


Figure 4.3: PIMS Service Portfolio Matrix (Centered Alignment)

4.1.4 PESTEL Analysis

To complement the internal SWOT analysis, a PESTEL framework is applied to evaluate the macro-environmental factors affecting PIMS Hospital's management and digital transformation.

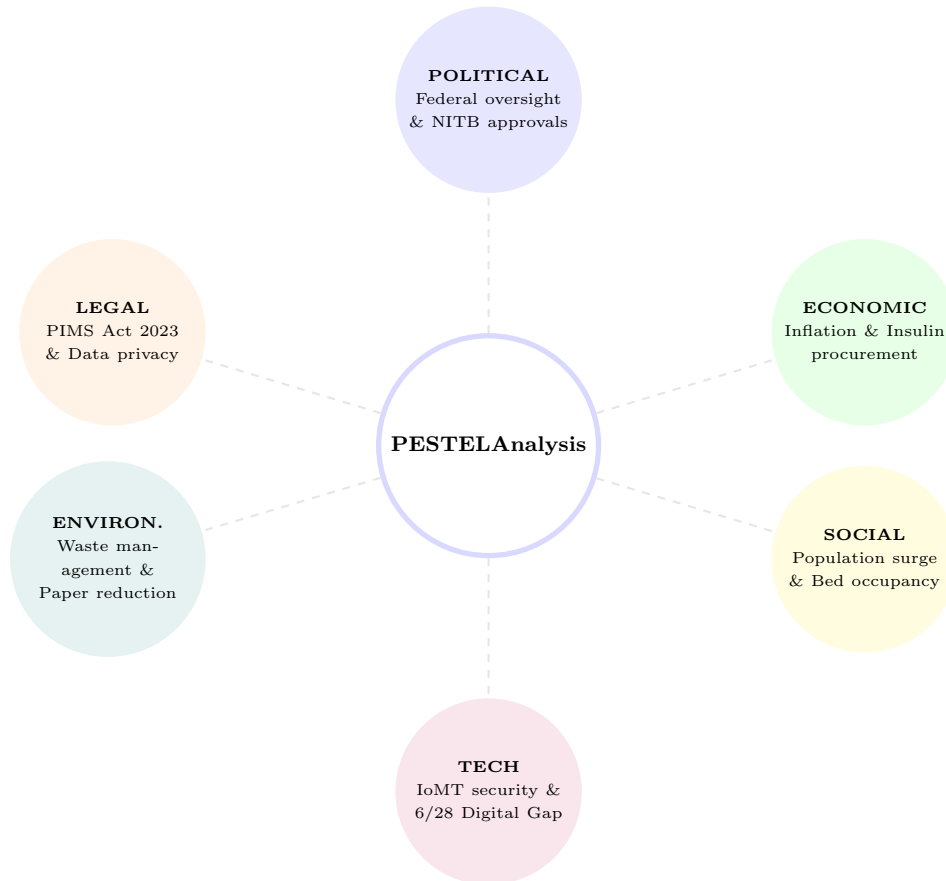


Figure 4.4: PESTEL Framework: Macro-Environmental Impact on PIMS

- **Political:** Federal pressure regarding the implementation of the 22 stalled digital modules and coordination with national IT boards.
- **Economic:** Impact of national inflation on procurement budgets, leading to shortages of critical medicines like Insulin.
- **Social:** High demand driven by a 73% bed occupancy rate, necessitating the "One Patient, One ID" digital system.
- **Technological:** Growth in IoMT connectivity (66.67%) and the transition from manual records to a unified digital ecosystem.
- **Environmental:** Efforts to reduce paper waste and optimize bio-medical waste disposal through digital tracking.

- **Legal:** Compliance with the PIMS Act 2023 and national regulations regarding electronic health record privacy.

4.2 Leadership and Team Management

Organizational performance is currently impacted by coordination failures between PIMS IT and external authorities. Utilizing **Maslow's Hierarchy of Needs**, it is analyzed that staff motivation is low due to a lack of operational "Safety" in project goals. We propose a **Democratic Leadership** approach to foster collaboration and resolve the internal conflict regarding system implementation.

4.3 Operations Management

Operational effectiveness is evaluated through the lens of resource optimization and logistics. By integrating the BI Dashboard's predictive insights, PIMS can implement **Just-in-Time (JIT)** inventory protocols for life-saving drugs. This transition from a reactive to a proactive workflow reduces "Process Waste" and ensures uninterrupted service delivery.

5. Integration of Technical and Management Skills

5.1 Technical and Managerial Integration

This section illustrates the fusion of management theories with technical Business Intelligence (BI) skills to address the organizational challenges at PIMS. The implementation of a Power BI dashboard serves as a bridge between raw operational data and strategic decision-making. By transforming fragmented IT data into actionable insights, management can move from a reactive state to proactive "Data-Driven Decision Making." This integration ensures that technical project completion is directly aligned with the hospital's overall operational performance.

5.2 Project Management Approach

To ensure the successful completion of the remaining 22 digital modules, a structured project management approach is required. Utilizing tools such as Gantt charts and Kanban boards allows for better tracking of resources and deadlines. This approach organizes the digital transformation into three distinct phases:

- **Phase 1 (Planning):** Addressing internal leadership conflicts and establishing clear technical milestones.
- **Phase 2 (Integration):** Synchronizing the Pharmacy and Laboratory modules with the central registration system to eliminate data silos.
- **Phase 3 (Deployment):** Full-scale rollout and staff training for the complete 28-module digital health suite.

5.2.1 Project Timeline: Gantt Chart

The following Gantt chart provides a visual representation of the project timeline for implementing the remaining 22 digital modules at PIMS.

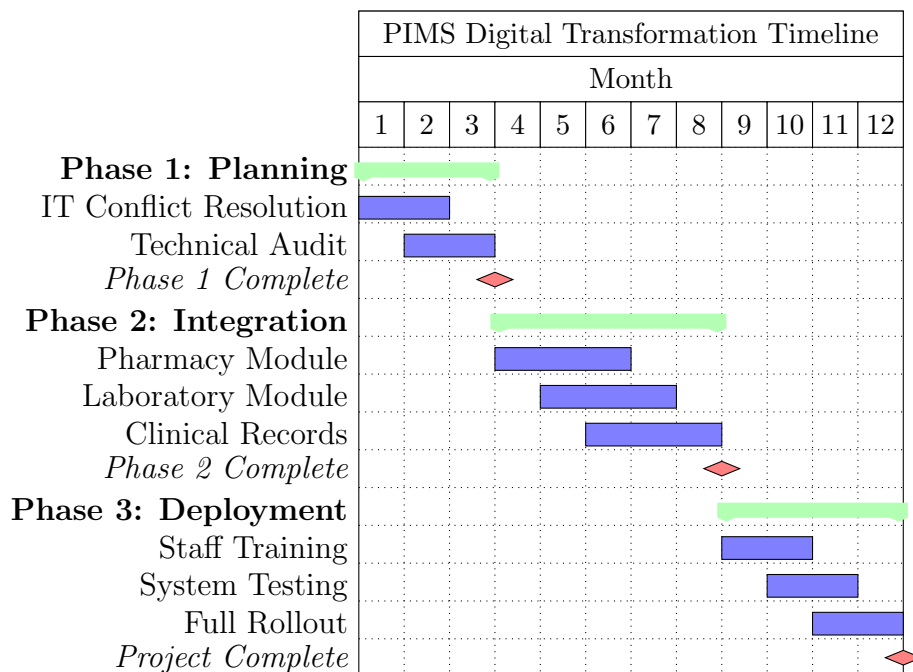


Figure 5.1: Project Management Gantt Chart for PIMS Digital Implementation

5.3 Risk Management

Effective risk management is vital for maintaining the operational effectiveness of a large-scale institution like PIMS. A Risk Matrix has been developed to prioritize and mitigate the technical and administrative threats identified during our process analysis.

Table 5.1: Aesthetic Risk Management Matrix for PIMS Hospital

Identified Risk	Impact	Probability	Mitigation Strategy
Digital Fragmentation	High	High	Accelerate module integration and adopt hybrid cloud solutions.
Supply Chain Failure	Critical	Medium	Implement automated inventory alerts and JIT protocols.
Internal Coordination Conflict	Medium	High	Establish a Joint Steering Committee and use Transformational Leadership.
Data Security Vulnerability	High	Medium	Utilize network segmentation and secure IoMT protocols.

5.4 Technical and Management Fusion

This analysis demonstrates how management fundamentals are integrated into technical projects. By using Business Intelligence to solve clinical bottlenecks, we illustrate how technical skills, such as data modeling, and management theories, such as resource optimization, work together. This fusion is essential for project success and long-term organizational stability in a digital-first healthcare environment.

6. Conclusion and Recommendations

6.1 Conclusion

This project has analyzed the key organizational processes of PIMS through the lens of Business Intelligence and Management Integration. The study revealed that while PIMS has a foundational digital infrastructure, it currently faces a significant "Digital Gap," with only 6 out of 28 planned modules being operational. This fragmentation has led to operational bottlenecks, specifically in the medicine supply chain and inter-departmental data coordination.

By utilizing Power BI dashboards, we have demonstrated how data-driven insights can provide the management visibility necessary to identify these inefficiencies and mitigate strategic risks. Ultimately, the integration of technical BI skills with management theories like SWOT and JIT inventory control offers a clear path toward enhancing hospital efficiency and patient safety.

6.2 Managerial Recommendations

To address the identified challenges and improve organizational performance, the following recommendations are proposed:

6.2.1 Short-term Improvements (0-6 Months)

- **Inventory Synchronization:** Immediately link the existing Pharmacy module with the Emergency and OPD consultation desks to provide real-time alerts for critical medicines like Insulin.
- **Conflict Resolution:** Establish a Joint Steering Committee between PIMS internal IT and external authorities to resolve leadership friction and set clear technical milestones.
- **Staff Training:** Conduct workshops for clinical staff on the "One Patient, One ID" system to ensure data entry is consistent and accurate at the point of care.

6.2.2 Medium-term Digital Integration (6-18 Months)

- **Module Acceleration:** Prioritize the activation of the remaining 22 modules, focusing first on the Laboratory and Radiology integration to eliminate physical "data carrying" by patients.
- **Cloud Migration:** Transition to a hybrid cloud environment to ensure that patient data is not "isolated" but accessible across federal health networks.
- **BI Expansion:** Expand the use of descriptive analytics to include financial and human resource management data for a complete organizational overview.

6.2.3 Long-term Strategic Vision (18+ Months)

- **Predictive Healthcare:** Implement Artificial Intelligence (AI) models to forecast patient inflow and disease outbreaks, moving PIMS from a reactive to a predictive management state.
- **IoMT Ecosystem:** Fully integrate the Internet of Medical Things (IoMT) to automate diagnostics and equipment monitoring, reducing human error and optimizing resource utilization.
- **Universal Health Record:** Lead the national initiative for a centralized health database, fulfilling the strategic vision of integrated and transparent digital health-care in Pakistan.

References

Pakistan Institute of Medical Sciences (PIMS). Official Portal. <https://pims.gov.pk>

Parliamentary Standing Committee on Health. (December 2025). *Report on Digital Transformation Implementation Gaps in Federal Tertiary Care Hospitals*. Dawn News. <https://www.dawn.com>

World Health Organization (WHO). *Digital Health and Medical Device Connectivity Standards*. <https://www.who.int>

National Information Technology Board (NITB). *National Digital Health Framework for Pakistan*. <https://nitb.gov.pk>

Robbins, S. P., & Coulter, M. *Management* (15th ed.). Pearson Education.

Porter, M. E. *Competitive Strategy: Techniques for Analyzing Industries and Competitors*. Free Press.

Microsoft. *Power BI Documentation: Decision Support Systems and Data Visualization*. <https://learn.microsoft.com/en-us/power-bi/>